



The algorithm and complexity of secure domination in 3-dimensional box graphs

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ABSTRACT

Given a graph G with vertex set V , a secure dominating set of G is a set $S \subseteq V$ with the property that for each vertex $u \in V \setminus S$, there is a vertex $v \in S$ adjacent to u such that $(S \cup \{u\}) \setminus \{v\}$ is a dominating set of G . The minimum secure dominating set (or, for short, MSDS) problem asks to find an MSDS in a given graph.

In this paper, firstly, we show that the MSDS problem is APX-hard in d -box graphs for any fixed integer $d \geq 2$. Secondly, we obtain a PTAS for the MSDS problem which is $(1 + \frac{21}{k})$ -approximation (resp. $(1 + \frac{6}{k})$ -approximation) and runs in $n^{O(k^3)}$ (resp. $n^{O(k^2)}$) time for unit cube graphs and unit ball graphs (resp. unit square graphs and unit disk graphs). Thirdly, we give a dynamic programming algorithm for the MSDS problem in unit-height box graphs (resp. unit-height rectangle graphs), where the centers of all boxes (resp. rectangles) are inside a column of length l and width w for some integers $l, w \geq 1$ (resp. a strip of width w for some integer $w \geq 1$). Finally, with the help of the dynamic programming algorithm, we improve the PTAS to $(1 + \frac{6}{k})$ -approximation (resp. $(1 + \frac{1}{k})$ -approximation) and $n^{O(k^2)}$ (resp. $n^{O(k)}$) time for unit cube graphs and unit ball graphs (resp. unit square graphs and unit disk graphs).

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1. Introduction

A graph G with vertex set V is called a *geometric intersection graph* if there is a bijection b between V and a set P of geometric objects such that there is an edge between two vertices u and v if and only if the objects $b(u)$ and $b(v)$ intersect. The set P is called a *geometric intersection representation* of G . A d -box is a subset of $\mathbb{R}^d$ that is a cartesian product of d intervals in $\mathbb{R}$. A d -box graph is a geometric intersection graph when P is a set of d -boxes. Clearly, a 2-box is a rectangle and a 2-box graph is also called a *rectangle graph*. A rectangle graph is *proper* if no rectangle is properly contained within another. If the width and height of a rectangle are w, h , respectively, then we say the rectangle has *size* $w \times h$, see Fig. 1(a). A *unit-height rectangle* (resp. *unit square*) is a rectangle with size $w \times 1$ and $w \geq 1$ (resp. 1×1). A 3-box (resp. 3-box graph) is abbreviated as a *box* (resp. *box graph*). Similarly, if the length, width and height of a box are l, w, h , respectively, then we say the box has *size* $l \times w \times h$, see Fig. 1(b). A *unit-height box* (resp. *unit cube*) is a box with size $l \times w \times 1$ and $w, l \geq 1$ (resp. $1 \times 1 \times 1$). If the height of a rectangle (resp. box) is unbounded, then we call the box is a *strip* (resp. *column*). When P is a set of unit-height rectangles (resp. unit squares, unit-height boxes, unit cubes, unit disks (the diameters of all disks are 1), unit balls (the diameters of all balls are 1)), the corresponding geometric intersection graph is called a *unit-height rectangle graph* (resp. *unit square graph*, *unit-height box graph*, *unit cube graph*, *unit disk graph*, *unit ball graph*). If the centers of all disks of a unit disk graph have integer coordinates, then the graph is called a *grid graph*. In the paper, all the d -boxes we mentioned are axis-parallel.

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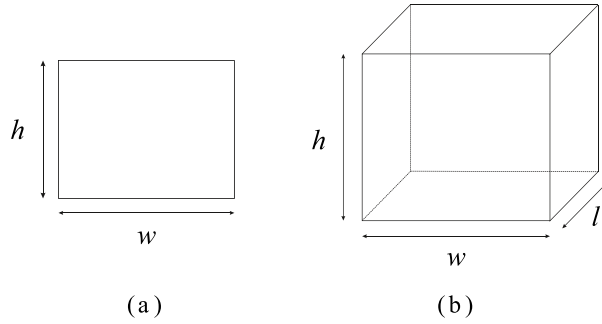


Fig. 1. (a) A rectangle has size $w \times h$. (b) A box has size $l \times w \times h$.

2-box graphs or rectangle graphs have been extensively studied as subclasses of geometric intersection graphs. Maximum-weight independent sets in rectangle graphs are widely studied due to their application in map labeling, VLSI design, and data mining [1,2,9]. Minimum dominating set was shown to be NP-hard even in diagonal-anchored rectangle graphs which are rectangle graphs when each rectangle is anchored at a line with slope -1 [15]. Bandyapadhyay [6] provided a $(2+\varepsilon)$ -approximation algorithm for the minimum dominating set in diagonal-anchored rectangle graphs. Wang et al. [18] studied minimum secure connected and secure total dominating sets in rectangle graphs. Chlebík et al. [10] proved some combinatorial optimization problems such as maximum independent set, minimum vertex cover, minimum dominating set, etc. are APX-hard in d -box graphs for any fixed $d \geq 3$.

Let U be a set of geometric objects and $G = (V, E)$ be a graph. For an object $u \in U$, let $N(u) = \{v \in U \setminus \{u\} : u \cap v \neq \emptyset\}$. Given two subsets W, U' of U with $N[W] \subseteq U'$, where $N[W] = \bigcup_{u \in W} N(u) \cup W$. A set $D \subseteq U'$ is a *dominating set of W from U'* (or, for short, a *DS of W from U'*) if $N(u) \cap D \neq \emptyset$ for each $u \in W \setminus D$. In particular, when $W = U' = U$, D is called a *DS of U* . In further, $D \subseteq V$ is a *DS of G* if every vertex not in D is adjacent to a vertex in D . A set $S \subseteq U'$ is a *secure dominating set of W from U'* (or, for short, an *SDS of W from U'*) if for each $u \in W \setminus S$, there is $v \in N(u) \cap S$ such that $(S \cup \{u\}) \setminus \{v\}$ is a DS of W from U' . In this case, we call v is an *S-defender of u* . Note that an SDS S of W from U' is also a DS of W from U' by the fact that each object $u \in W \setminus S$ is dominated by an S -defender of u . In particular, when $W = U' = U$, S is referred as an *SDS of U* . In further, a set $S \subseteq V$ is an *SDS of G* if for each vertex $x \in V \setminus S$, there is a vertex $y \in S$ adjacent to x such that $(S \cup \{x\}) \setminus \{y\}$ is a DS of G . The *secure domination number* $\gamma_s(G)$ is the minimum cardinality among all SDSs of G . If the set U' is clear from the context, then we omit the term “from U' ” and simply say “a DS of W ” and “an SDS of W ”.

Secure domination was introduced by Cockayne et al. in [11] and has received a lot of research. The *MSDS problem* is the problem of finding a minimum SDS in a given graph. The problem is linear-time solvable in block graphs [16], cographs [5,13] and proper interval graphs [4]. Burger et al. [8] gave two exponential-time algorithms for the MSDS problem in general graphs. Wang et al. [19] showed that the MSDS problem cannot be approximated within a factor of $(1 - \varepsilon) \ln |V|$ for any $\varepsilon > 0$, unless $\text{NP} \subseteq \text{DTIME}(|V|^{O(\log \log |V|)})$ and the MSDS problem is APX-complete for bounded degree graphs. The MSDS problem was shown to be NP-hard in bipartite graphs and split graphs [14], star convex bipartite graphs and doubly chordal graphs [19], undirected path graphs and chordal bipartite graphs [16], and grid graph [17]. Since grid graphs are a subclass of unit square graphs [7] and unit disk graphs, the MSDS problem is NP-hard in unit square graphs and unit disk graphs, also in unit cube graphs and unit ball graphs.

The paper is organized as follows. In Section 2, we show that the MSDS problem is APX-hard in d -box graphs for any fixed integer $d \geq 2$. In Section 3, we propose PTASs for the MSDS problem in unit cube graphs, unit ball graphs, unit square graphs and unit disk graphs. In detail, in Section 3.1, we obtain a PTAS which is $(1 + \frac{21}{k})$ -approximation (resp. $(1 + \frac{6}{k})$ -approximation) and runs in $n^{O(k^3)}$ (resp. $n^{O(k^2)}$) time for unit cube graphs and unit ball graphs (resp. unit square graphs and unit disk graphs). In Section 3.2, we give a dynamic programming algorithm for the MSDS problem in unit-height box graphs (resp. unit-height rectangle graphs), where the centers of all boxes (resp. rectangles) are inside a column of length l and width w for some integers $l, w \geq 1$ (resp. a strip of width w for some integer $w \geq 1$). In Section 3.3, with the help of the dynamic programming algorithm, we improve the PTAS to $(1 + \frac{6}{k})$ -approximation (resp. $(1 + \frac{1}{k})$ -approximation) and $n^{O(k^2)}$ (resp. $n^{O(k)}$) time for unit cube graphs and unit ball graphs (resp. unit square graphs and unit disk graphs).

2. APX-hardness

In this section, we show that the MSDS problem is APX-hard in d -box graphs for any fixed integer $d \geq 2$. Thus the MSDS problem has no PTAS in d -box graphs for any fixed integer $d \geq 2$.

We first discuss the case of 2-box graphs, that is, rectangle graphs. Given a graph $G = (V, E)$, a *vertex cover* of G is a set $C \subseteq V$ such that each edge $uv \in E$, either $u \in C$ or $v \in C$. The *vertex cover number* $\beta(G)$ is the minimum cardinality

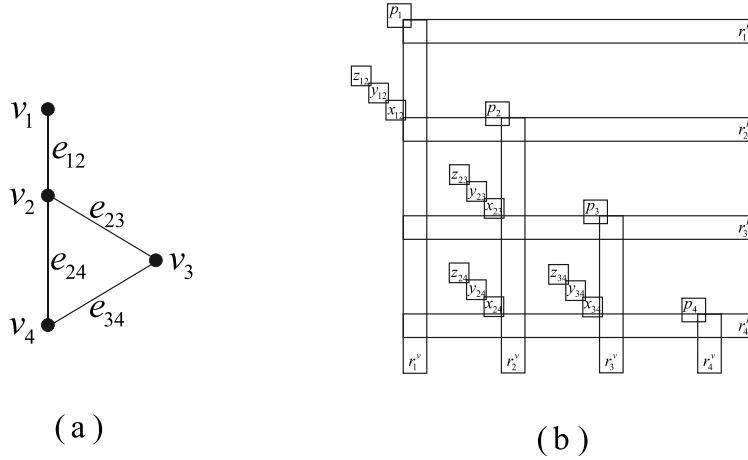


Fig. 2. (a) A graph G with maximum degree 3. (b) A proper rectangle graph G' obtained from G .

among all vertex covers of G . The *minimum vertex cover problem* is a problem of finding a minimum vertex cover in a given graph.

Theorem 2.1. *The MSDS problem is APX-hard in proper rectangle graphs.*

Proof. We offer an L-reduction from the APX-hard problem [3] minimum vertex cover in a graph $G = (\{v_1, \dots, v_n\}, E)$ with maximum degree at most 3 to the MSDS problem in a proper rectangle graph G' . Let $|E| = m$. For each edge $e_{ij} \in E$, suppose $i < j$ and denote $e_{ij} = v_i v_j$. A proper rectangle graph G' is constructed as follows:

Step 1. For each vertex v_i , construct a thin horizontal rectangle r_i^h , a thin vertical rectangle r_i^v and a little square p_i . On the whole, place $r_1^h, \dots, r_n^h$ from top to bottom that do not intersect each other such that their left boundaries are on a vertical line l . Place $r_1^v, \dots, r_n^v$ from left to right that do not intersect each other such that: (1) the left boundary of r_1^v is on the vertical line l , (2) the top boundary of r_i^v is on the top boundary of r_i^h for any $1 \leq i \leq n$. Then for any $1 \leq i \leq n$, r_i^v and r_i^h have an intersection region R_i . Place p_i such that p_i intersects with R_i .

Step 2. For r_i^v , there is an intersection region S_{ij} with r_j^h , where $1 \leq i < j \leq n$. For each $e_{ij} \in E$, construct three same little squares x_{ij}, y_{ij}, z_{ij} such that z_{ij} intersects y_{ij} , y_{ij} intersects x_{ij} , and x_{ij} intersects y_{ij} and S_{ij} .

Step 3. Adjusting the size and position of rectangles appropriately will result in rectangles not containing each other. Fig. 2(b) is an example of the construction.

Given a minimum vertex cover C of G . Let $S = \{r_i^h, r_i^v : v_i \in C\} \cup \{p_i : v_i \notin C\} \cup \{y_{ij} : e_{ij} \in E\}$. Next we show that S is an SDS of G' . For each $1 \leq i \leq n$, if $p_i \notin S$, then $r_i^h, r_i^v \in S$, it follows that r_i^h is an S -defender of p_i . Otherwise $p_i \in S$, then $r_i^h, r_i^v \notin S$, it follows that p_i is an S -defender of r_i^h and r_i^v . Since C is a vertex cover of G , there is $v_i \in C$ or $v_j \in C$ for each $e_{ij} \in E$, that is, $r_i^h, r_i^v \in S$ or $r_j^h, r_j^v \in S$. Thus for each z_{ij}, y_{ij} is an S -defender of z_{ij} . For each x_{ij} , if $r_i^h, r_i^v \in S$, then r_i^v is an S -defender of x_{ij} , otherwise r_j^h is an S -defender of x_{ij} . Since the maximum degree of G is 3, $\beta(G) \geq n/4$ and $m \leq 3n/2$. Hence $\gamma_s(G') \leq |S| = |C| + n + m = \beta(G) + n + m \leq 11\beta(G)$.

Given an SDS S of G' . We will construct an SDS S' of G' from S satisfying the following conditions: (1) $|S'| \leq |S|$; (2) for each $e_{ij} \in E$, $z_{ij}, x_{ij} \notin S'$ and $y_{ij} \in S'$; (3) $S' \cap \{r_i^h, p_i, r_i^v\} = \{r_i^h, r_i^v\}$ or $|S' \cap \{r_i^h, p_i, r_i^v\}| = 1$ for each $1 \leq i \leq n$. For each $e_{ij} \in E$, since z_{ij} needs to be dominated by S , $|S \cap \{y_{ij}, z_{ij}\}| \geq 1$. If $z_{ij} \in S$, $y_{ij} \notin S$, then let $S_1 = (S \setminus \{z_{ij}\}) \cup \{y_{ij}\}$; otherwise, $z_{ij}, y_{ij} \in S$, then let $S_1 = (S \setminus \{z_{ij}\}) \cup \{x_{ij}\}$. In both cases, S_1 is an SDS of G' with size at most $|S|$ and $z_{ij} \notin S_1$, $y_{ij} \in S_1$ for each $e_{ij} \in E$. For each $x_{ij} \in S_1$, if $r_i^h, r_j^h \in S_1$, then let $S_2 = (S_1 \setminus \{x_{ij}\}) \cup \{r_j^v\}$; otherwise, $r_i^h \notin S_1$ or $r_j^h \notin S_1$, without loss of generality, suppose $r_j^h \notin S_1$, then let $S_2 = (S_1 \setminus \{x_{ij}\}) \cup \{r_j^h\}$. In both cases, r_j^h is an S_2 -defender of x_{ij} . Then S_2 is an SDS of G' with $|S_2| \leq |S_1| \leq |S|$ and $x_{ij} \notin S_2$ for any $e_{ij} \in E$. For each $1 \leq i \leq n$, since p_i needs to be dominated by S_2 , $|S_2 \cap \{r_i^h, p_i, r_i^v\}| \geq 1$. If $|S_2 \cap \{r_i^h, p_i, r_i^v\}| \geq 2$, then let $S' = (S_2 \setminus \{r_i^h, p_i, r_i^v\}) \cup \{r_i^h, r_i^v\}$. Obviously, S' is an SDS of G' satisfying the conditions (1), (2) and (3).

Let $C = \{v_i : r_i^v, r_i^h \in S'\}$. For each $e_{ij} \in E$, y_{ij} is not an S' -defender of x_{ij} , then an S' -defender of x_{ij} is r_i^v or r_j^h . Without loss of generality, suppose r_j^h is an S' -defender of x_{ij} , that is, $(S' \cup \{x_{ij}\}) \setminus \{r_j^h\}$ is a DS of G' . Since p_j needs to be dominated by $(S' \cup \{x_{ij}\}) \setminus \{r_j^h\}$, $p_j \in S'$ or $r_j^v \in S'$. Then $|S' \cap \{r_j^h, p_j, r_j^v\}| \geq 2$. Thus $r_j^h, r_j^v \in S'$ by condition (3). Then $v_j \in C$, it follows that C is a vertex cover of G . Due to conditions (1), (2) and (3), $|S| \geq |S'| = |C| + n + m$. If S is a minimum SDS of G' , then $\gamma_s(G') = |S| \geq |C| + n + m \geq \beta(G) + n + m$. Hence $\gamma_s(G') = \beta(G) + n + m$. Thus $|S| - \gamma_s(G') \geq |C| - \beta(G)$. $\square$

By Theorem 2.1 and proper rectangle graphs are a subclass of rectangle graphs, we have the following result.

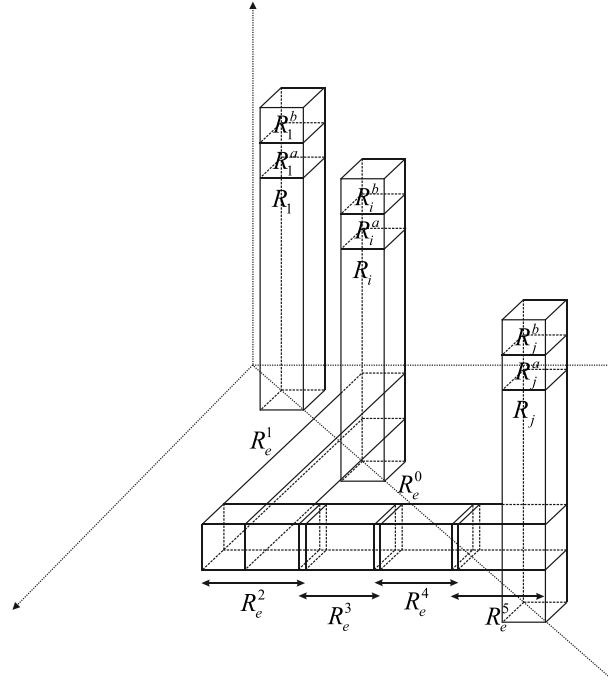


Fig. 3. An illustration for Corollary 2.4.

Corollary 2.2. *The MSDS problem is APX-hard in rectangle graphs.*

Now we consider the case of $d \geq 3$. For a graph G and an integer $k \geq 0$, a k -subdivision of an edge $e = uv$ in G is defined as a replacement of e by a path with endvertices u and v and with k new internal vertices.

Theorem 2.3 ([10]). *Let $G = (V, E)$ be a graph and $s(e) \geq 2$ be an integer for each edge $e \in E$. Denote by G' a graph obtained from G by an $s(e)$ -subdivision of each edge $e \in E$. Then for any fixed integer $d \geq 3$, the graph G' can be realized as a d -box graph. Moreover, such realization can be done in time polynomial in $|V| + \sum_{e \in E} s(e)$.*

By Theorem 2.3, the following result can be easily obtained.

Corollary 2.4. *Let $G = (V, E)$ be a graph and $s(e) \geq 2$ be an integer for each edge $e \in E$. Denote by $G' = (V', E')$ a graph obtained from G by an $s(e)$ -subdivision of each edge $e \in E$. Define a graph H as vertex set $V(H) = V' \cup \{v^a, v^b : v \in V\}$ and edge set $E(H) = E' \cup \{vv^a, v^av^b : v \in V\}$. Then for any fixed integer $d \geq 3$, the graph H can be realized as a d -box graph. Moreover, such realization can be done in time polynomial in $3|V| + \sum_{e \in E} s(e)$.*

Proof. On the basis of the proof of Theorem 2.3, firstly, extend the box R_i to $[2i - 1, 2i] \times [2i - 1, 2i] \times [1, 2|E| + 1]$. Secondly, for each v_i^a , construct a box $R_i^a = [2i - 1, 2i] \times [2i - 1, 2i] \times [2|E| + 1, 2|E| + 2]$, for each v_i^b , construct a box $R_i^b = [2i - 1, 2i] \times [2i - 1, 2i] \times [2|E| + 2, 2|E| + 3]$. The remaining proof is the same as Theorem 2.3. See Fig. 3 for an illustration. $\square$

Theorem 2.5. *The MSDS problem is APX-hard in d -box graphs, where $d \geq 3$.*

Proof. We offer an L-reduction from the APX-hard problem [3] minimum dominating set in a graph $G = (\{v_1, \dots, v_n\}, E)$ with maximum degree at most 3 to the MSDS problem in a d -box graph H . Now we construct the d -box graph H . Let $|E| = m$. First construct a graph $G' = (V', E')$ which is obtained from G by a 7-subdivision of each edge $e \in E$. Clearly, each edge $v_i v_j$ in G corresponds a path $v_i s_{ij}^1 \dots s_{ij}^7 v_j$ in G' . Then define the graph H as vertex set $V(H) = V' \cup \{v_i^a, v_i^b : 1 \leq i \leq n\}$ and edge set $E(H) = E' \cup \{v_i v_i^a, v_i^a v_i^b : 1 \leq i \leq n\}$. By Corollary 2.4, H is a d -box graph and the construction can be realized in polynomial time.

Given a minimum DS D of G . We will construct an SDS S of H . Firstly, add D and v_i^a for all $1 \leq i \leq n$ into S . Secondly, for each edge $v_i v_j$ in G , if $v_i \in D$, then add $s_{ij}^2, s_{ij}^5, s_{ij}^7$ into S . The case $v_j \in D$ can be considered similarly. Otherwise $v_i, v_j \notin D$, then add $s_{ij}^2, s_{ij}^4, s_{ij}^6$ into S . Now we show that S is an SDS of H . Clearly, S is a DS of H . For each v_i^b , v_i^a is an

S -defender of v_i^b . For each $v_j \notin S$, there is $v_i \in D \subseteq S$ such that $v_i v_j \in E$. Then s_{ij}^7 is an S -defender of v_j . For each edge $v_i v_j \in E$, first we consider the case of $v_i \in S$, then v_i (resp. $s_{ij}^2, s_{ij}^5, s_{ij}^7$) is an S -defender of s_{ij}^1 (resp. $s_{ij}^3, s_{ij}^4, s_{ij}^6$). The case $v_j \in S$ can be considered similarly. Second we consider the case of $v_i, v_j \notin S$, then s_{ij}^2 (resp. s_{ij}^4, s_{ij}^6) is an S -defender of s_{ij}^1 (resp. $s_{ij}^3, s_{ij}^5, s_{ij}^7$). Therefore S is an SDS of H . Since the maximum degree of G is 3, $\gamma(G) \geq n/4$ and $m \leq 3n/2$. Then $\gamma_s(H) \leq |S| = |D| + n + 3m = \gamma(G) + n + 3m \leq 23\gamma(G)$.

Given an SDS S of H . We will construct an SDS S' of H from S satisfying the following conditions: (1) $|S'| \leq |S|$; (2) for each $1 \leq i \leq n$, $v_i^b \notin S'$; (3) $|S' \cap P_{ij}| = 3$ for each $v_i v_j \in E$, where $P_{ij} = \{s_{ij}^1, \dots, s_{ij}^7\}$. For each $1 \leq i \leq n$, if $v_i^a, v_i^b \in S$, then let $S_1 = (S \cup \{v_i\}) \setminus \{v_i^b\}$. If $v_i^b \in S$ and $v_i^a \notin S$, then let $S_1 = (S \cup \{v_i^a\}) \setminus \{v_i^b\}$. Clearly, $|S_1| \leq |S|$ and S_1 is an SDS of H with $v_i^b \notin S_1$ and $v_i^a \in S_1$.

Now we show that $|S_1 \cap P_{ij}| \geq 3$ for each $v_i v_j \in E$. Clearly, $|S_1 \cap P_{ij}| \geq 2$. Suppose to the contrary that $|S_1 \cap P_{ij}| = 2$. Clearly $s_{ij}^1, s_{ij}^4, s_{ij}^7 \notin S_1$, otherwise, at least one vertex in P_{ij} is not dominated by S_1 . If $s_{ij}^2 \in S_1$, then $s_{ij}^5 \in S_1$ and there is no S_1 -defender of s_{ij}^4 , a contradiction. So $s_{ij}^2, s_{ij}^6 \notin S_1$ by the symmetry. Then $s_{ij}^3, s_{ij}^5 \in S_1$, but there is no S_1 -defender of s_{ij}^4 , a contradiction. Therefore $|S_1 \cap P_{ij}| \geq 3$ for each $v_i v_j \in E$. For each $v_i v_j \in E$ with $|S_1 \cap P_{ij}| \geq 4$, if $v_i, v_j \in S_1$, then let $S' = (S_1 \setminus P_{ij}) \cup \{s_{ij}^2, s_{ij}^4, s_{ij}^6\}$. Otherwise $v_i \notin S_1$ or $v_j \notin S_1$, without loss of generality, assume $v_i \notin S_1$, then let $S' = (S_1 \setminus P_{ij}) \cup \{v_i, s_{ij}^2, s_{ij}^5, s_{ij}^7\}$. Thus we obtain a set S' which is an SDS of H and satisfies the conditions (1), (2) and (3).

Next we show that $D = \{v_i \in V : v_i \in S'\}$ is a DS of G . For any $v_j \notin D$, then $v_j \notin S'$. Since S' is an SDS of H and $v_j^b \notin S'$, v_j^a is not an S' -defender of v_j and further there is an S' -defender s_{ij}^7 of v_j , where $v_i v_j \in E$ and $s_{ij}^7 \in P_{ij}$. By condition (3), either $S' \cap P_{ij} = \{s_{ij}^3, s_{ij}^5, s_{ij}^7\}$ or $S' \cap P_{ij} = \{s_{ij}^2, s_{ij}^5, s_{ij}^7\}$. By simple analysis, v_i is an S' -defender of s_{ij}^1 in either case. So $v_i \in S'$ and further $v_i \in D$. Thus D is a DS of G . By conditions (1), (2) and (3), $|S| \geq |S'| = |D| + n + 3m$. If S is a minimum SDS of H , then $\gamma_s(H) = |S| \geq |D| + n + 3m \geq \gamma(G) + n + 3m$. Hence $\gamma_s(H) = \gamma(G) + n + 3m$. Thus $|S| - \gamma_s(H) \geq |D| - \gamma(G)$. $\square$

3. PTAS

By the above results, the MSDS problem has no PTAS in rectangle graphs and box graphs. But for some restricted rectangle graphs and box graphs, PTAS exists. In the following we present PTAS for the MSDS problem in unit square graphs, unit disk graphs, unit cube graphs and unit ball graphs. The PTAS is designed by using the shifting strategy which was introduced by Hochbaum and Maass [12]. We use unit cube graphs as an example to discuss the algorithm, the other graph classes can be considered similarly. Firstly, we apply 3-level shifting strategy to obtain a PTAS which is $(1 + \frac{21}{k})$ -approximation and runs in $n^{O(k^3)}$ time. Then we give a dynamic programming algorithm for the MSDS problem in unit-height box graphs, where centers of all boxes are inside a column of length l and width w for some integers $l, w \geq 1$. Thirdly, with the help of the dynamic programming algorithm, using 2-level shifting strategy, the PTAS can be improved to $(1 + \frac{6}{k})$ -approximation and $n^{O(k^2)}$ time.

3.1. 3-Level shifting strategy

In this subsection, we apply 3-level shifting strategy to discuss PTAS for the MSDS problem. We only discuss the case of unit cube graphs, the other graph classes are similar.

Given a set U of n unit cubes. Let $C = L \times W \times H$ be a smallest box containing centers of all cubes in U and $k \geq 1$ be an integer. Suppose the left-bottom-hind corner of C is origin $(0, 0, 0)$. Define a xoz -plane is a plane which has the same normal vector as the xoz plane. Similarly, we can define a yoz -plane and xoy -plane.

The shifting strategy contains 3 levels. The first level contains k iterations. In the i th iteration ($1 \leq i \leq k$), we use xoz -planes $y = 6i, y = 6i + 6k, \dots, y = 6i + 6ka_i$ ($a_i \geq 0$ is an integer) to partition C into smaller boxes such that the first box has size $L \times 6i \times H$ and the remaining boxes have size $L \times 6k \times H$ (the size of the last box may less than $L \times 6k \times H$), see Fig. 4.

We consider any one of the boxes C_y with size $L \times 6k \times H$ (the box with size less than $L \times 6k \times H$ can be considered similarly). The second level contains k iterations. In the j th iteration ($1 \leq j \leq k$), we use yoz -planes $x = 6j, x = 6j + 6k, \dots, x = 6j + 6kb_j$ ($b_j \geq 0$ is an integer) to partition C_y into smaller boxes such that the first box has size $6j \times 6k \times H$ and the remaining boxes have size $6k \times 6k \times H$ (the size of the last box may less than $6k \times 6k \times H$).

We consider any one of the boxes C_{yx} with size $6k \times 6k \times H$ (the box with size less than $6k \times 6k \times H$ can be considered similarly). The third level contains k iterations. In the l th iteration ($1 \leq l \leq k$), we use xoy -planes $z = 6l, z = 6l + 6k, \dots, z = 6l + 6kc_l$ ($c_l \geq 0$ is an integer) to partition C_{yx} into smaller boxes such that the first box has size $6k \times 6k \times 6l$ and the remaining boxes have size $6k \times 6k \times 6k$ (the size of the last box may less than $6k \times 6k \times 6k$).

Let C be any one of the boxes with size $6k \times 6k \times 6k$ (the boxes with size less than $6k \times 6k \times 6k$ can be considered similarly). For an integer $d \geq 0$, expanding C by d units will obtain a box C^d which has the same center with C and has size $(6k + 2d) \times (6k + 2d) \times (6k + 2d)$. Note that $C^0 = C$. Suppose the top, right and front sides of C^d are open except the sides of C , that is, if a point is on the top, right and front sides of C^d , then the point is not inside C^d . Let $U_{C^d} \subseteq U$ be the set of cubes whose centers are contained in C^d . Define OPT as a minimum SDS of U and OPT_{C^d} as a minimum SDS of U_{C^d} from U_{C^3} .

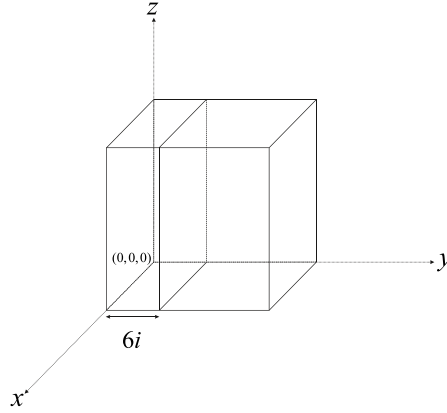


Fig. 4. A partition using xoz-plane $y = 6i$.

Observation 3.1. Let u, u' be any two unit cubes in U with centers c, c' , respectively. If c and c' are contained in a unit cube, then $u \cap u' \neq \emptyset$.

Lemma 3.2. OPT_{C^2} can be obtained in time $n_{C^3}^{O(k^3)}$, where $n_{C^3} = |U_{C^3}|$.

Proof. Partition C^2 into cells of size $1 \times 1 \times 1$. Then C^2 contains $(6k+4)^3 = O(k^3)$ cells. Suppose the top, right and front sides of each cell are open. If the center of a cube is in a cell, then we simply say the cube is in the cell. For each cell, pick any cube in the cell (if it exists) into a set S_{C^2} . By Observation 3.1, any two cubes in a cell are intersecting. So S_{C^2} is an SDS of U_{C^2} . Thus $|OPT_{C^2}| \leq |S_{C^2}| \leq O(k^3)$. Then OPT_{C^2} can be obtained by checking all subsets in U_{C^3} with size at most $O(k^3)$, which takes $n_{C^3}^{O(k^3)}$ time. $\square$

In the following we consider any (i, j, l) -th iteration, where $1 \leq i, j, l \leq k$. Suppose there are s boxes $C_1, \dots, C_s$ in this partition. Let $S_{i,j,l} = OPT_{C_1^2} \cup \dots \cup OPT_{C_s^2}$. The algorithm outputs a set S , which has minimum cardinality among all $\{S_{i,j,l} : 1 \leq i, j, l \leq k\}$, that is, $|S| = \min_{1 \leq i, j, l \leq k} |S_{i,j,l}|$.

Lemma 3.3. S is an SDS of U .

Proof. We only need to show that $S_{i,j,l}$ is an SDS of U , for any $1 \leq i, j, l \leq k$. Since $U = U_{C_1^2} \cup \dots \cup U_{C_s^2}$ and $OPT_{C_t^2}$ is a DS of $U_{C_t^2}$ for any $1 \leq t \leq s$, $S_{i,j,l}$ is a DS of U .

For any $u \in U \setminus S_{i,j,l}$, there is unique $1 \leq t \leq s$ such that $u \in U_{C_t^2}$. Then $u \in U_{C_t^2}$ and $u \notin OPT_{C_t^2}$ by $u \notin S_{i,j,l}$. Since $OPT_{C_t^2}$ is an SDS of $U_{C_t^2}$, there is $u' \in OPT_{C_t^2} \cap N(u)$ such that $(OPT_{C_t^2} \cup \{u\}) \setminus \{u'\}$ is a DS of $U_{C_t^2}$. Now we show that $S'_{i,j,l} = (S_{i,j,l} \cup \{u\}) \setminus \{u'\}$ is a DS of U . Since $S_{i,j,l}$ is a DS of U , we only need to consider any cube $v \in N(u') \cap (U \setminus S'_{i,j,l})$. Since $u \in U_{C_t^2}$, $u' \in U_{C_t^2}$, further $v \in U_{C_t^2}$. Since $(OPT_{C_t^2} \cup \{u\}) \setminus \{u'\}$ is a DS of $U_{C_t^2}$, there is $v' \in ((OPT_{C_t^2} \cup \{u\}) \setminus \{u'\}) \cap N(v)$. Clearly, $v' \in S'_{i,j,l}$. Thus $S'_{i,j,l}$ is a DS of U . $\square$

Lemma 3.4. $|S| \leq (1 + \frac{21}{k})|OPT|$.

Proof. Consider any (i, j, l) -th iteration, where $1 \leq i, j, l \leq k$. Let OPT_i^a (resp. OPT_j^b, OPT_l^c) be the set of cubes in OPT whose centers are contained in the box $L \times [6i+6ka-3, 6i+6ka+3) \times H$ for any $0 \leq a \leq a_i$ (resp. $[6j+6kb-3, 6j+6kb+3) \times W \times H$ for any $0 \leq b \leq b_j, L \times W \times [6l+6lc-3, 6l+6lc+3)$ for any $0 \leq c \leq c_l$). Let $OPT_i^Y = \bigcup_{a=0}^{a_i} OPT_i^a, OPT_j^X = \bigcup_{b=0}^{b_j} OPT_j^b, OPT_l^Z = \bigcup_{c=0}^{c_l} OPT_l^c$ and $OPT_{i,j,l} = OPT_i^Y \cup OPT_j^X \cup OPT_l^Z$. Note that for any $1 \leq i \neq i' \leq k$, a cube cannot be in both OPT_i^Y and $OPT_{i'}^Y$, that is, $OPT_i^Y \cap OPT_{i'}^Y = \emptyset$. Then

$$\sum_{i=1}^k |OPT_i^Y| = |\bigcup_{i=1}^k OPT_i^Y| \leq |OPT|. \quad (3.1)$$

Similarly, we have

$$\sum_{j=1}^k |OPT_j^X| = |\bigcup_{j=1}^k OPT_j^X| \leq |OPT|. \quad (3.2)$$

$$\sum_{l=1}^k |OPT_l^Z| = |\bigcup_{l=1}^k OPT_l^Z| \leq |OPT|. \quad (3.3)$$

By inequalities (3.1), (3.2) and (3.3), we have

$$\sum_{i=1}^k \sum_{j=1}^k \sum_{l=1}^k |OPT_{i,j,l}| \leq k^2 \sum_{i=1}^k |OPT_i^Y| + k^2 \sum_{j=1}^k |OPT_j^X| + k^2 \sum_{l=1}^k |OPT_l^Z| \leq 3k^2 |OPT|. \quad (3.4)$$

For any $1 \leq t \leq s$, $OPT_{C_t^2}$ is a minimum SDS of $U_{C_t^2}$. Since $OPT \cap U_{C_t^3}$ is an SDS of $U_{C_t^2}$, then

$$|OPT_{C_t^2}| \leq |OPT \cap U_{C_t^3}|. \quad (3.5)$$

Thus

$$\begin{aligned} |S_{i,j,l}| &\leq |OPT_{C_1^2}| + \cdots + |OPT_{C_s^2}| \\ &\leq |OPT \cap U_{C_1^3}| + \cdots + |OPT \cap U_{C_s^3}| \\ &\leq |OPT| + 7|OPT_{i,j,l}|. \end{aligned} \quad (3.6)$$

The second inequality is obtained from (3.5). The third inequality is obtained by the fact that only the cubes in $OPT_{i,j,l}$ can be repetitively calculated and each cube is repetitively calculated at most 8 times.

Therefore by inequalities (3.4) and (3.6) we have

$$\begin{aligned} |S| &= \min_{1 \leq i,j,l \leq k} |S_{i,j,l}| \\ &\leq \frac{1}{k^3} \sum_{i=1}^k \sum_{j=1}^k \sum_{l=1}^k |S_{i,j,l}| \\ &\leq \frac{1}{k^3} (k^3 |OPT| + 7 \sum_{i=1}^k \sum_{j=1}^k \sum_{l=1}^k |OPT_{i,j,l}|) \\ &\leq \frac{1}{k^3} (k^3 |OPT| + 21k^2 |OPT|) \\ &= (1 + \frac{21}{k}) |OPT|. \quad \square \end{aligned}$$

Theorem 3.5. Given a set U of n unit cubes and an integer $k \geq 1$. The proposed 3-level shifting algorithm produces an SDS of U with size at most $(1 + \frac{21}{k})|OPT|$ in time $n^{O(k^3)}$.

Proof. By Lemmas 3.3, 3.4, the algorithm produces an SDS of U with size at most $(1 + \frac{21}{k})|OPT|$. Now we discuss the running time. For each (i, j, l) -th iteration ($1 \leq i, j, l \leq k$), the time taken to compute OPT_{C^2} is $n_{C^3}^{O(k^3)}$ by Lemma 3.2, where C is any one of the boxes in this iteration and $n_{C^3} = |U_{C^3}|$. Since each cube participates in computing $S_{i,j,l}$ at most 8 boxes, computing $S_{i,j,l}$ needs $n^{O(k^3)}$ time. There are k^3 iterations, the total running time is $n^{O(k^3)}$. $\square$

When U is a set of unit balls, replace the cell of size $1 \times 1 \times 1$ in Lemma 3.2 with the cell of size $\frac{1}{\sqrt{3}} \times \frac{1}{\sqrt{3}} \times \frac{1}{\sqrt{3}}$. Then any two unit balls with centers in a cell intersect. Thus the 3-level shifting algorithm is also effective for unit ball graphs. Hence we have the following result.

Corollary 3.6. Given a set U of n unit balls and an integer $k \geq 1$. The proposed 3-level shifting algorithm produces an SDS of U with size at most $(1 + \frac{21}{k})|OPT|$ in time $n^{O(k^3)}$.

For 2-dimensional situation, 3-level shifting strategy can be reduced to 2 levels. Then there is a better result.

Corollary 3.7. Let U be a set of n unit squares or n unit disks and $k \geq 1$ be an integer. Then there is a 2-level shifting algorithm that produces an SDS of U with size at most $(1 + \frac{6}{k})|OPT|$ in time $n^{O(k^2)}$.

3.2. The (l, w) -column MSDS problem

Let U be a set of n unit-height boxes whose centers are inside a column T of length l and width w for some integers $l, w \geq 1$, and W be a subset of U . In this subsection, we describe a dynamic programming approach to find a minimum

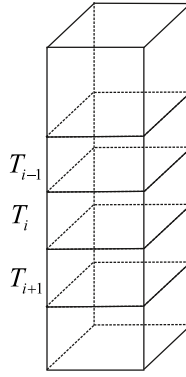


Fig. 5. Three $(l, w, 1)$ -boxes T_{i-1}, T_i, T_{i+1} .

SDS of W from U .

Firstly, we use xoy -plane to partition the column T into $(l, w, 1)$ -boxes which are boxes with size $l \times w \times 1$, see Fig. 5. Suppose the top side of each $(l, w, 1)$ -box is open. Secondly, we remove all the $(l, w, 1)$ -boxes that do not contain any centers of boxes in U . Let $T_1, \dots, T_s$ be the remaining $(l, w, 1)$ -boxes which are ordered from top to bottom. Obviously, $s \leq n$. For any $1 \leq i \leq s$, let $U_i \subseteq U$ be the set of boxes whose centers are inside T_i and $W_i = W \cap U_i$. Let $U_0 = U_{s+1} = \dots = U_{s+5} = W_{s+1} = \dots = W_{s+4} = \emptyset$.

Observation 3.8. Let $u \in U_i$ and any $u' \in N(u)$, where $1 \leq i \leq s$. Then $u' \in U_{i-1} \cup U_i \cup U_{i+1}$.

Lemma 3.9. For any minimum SDS D of W from U , $|D \cap U_i| \leq 3lw$ for any $1 \leq i \leq s$.

Proof. Partition the $(l, w, 1)$ -box T_i into cells of size $1 \times 1 \times 1$ for any $1 \leq i \leq s$. Then there are lw cells in T_i . Suppose the top, right and front sides of a cell are open except the sides of T . If the center of a box is in a cell, then we simply say the box is in the cell. By Observation 3.1, any two boxes in a cell are intersecting.

Suppose to the contrary that there is a minimum SDS D' of W such that $|D' \cap U_i| > 3lw$ for some $1 \leq i \leq s$. For each cell of T_i , select one box in the cell if it exists. For each cell of $T_{i-1} \cup T_{i+1}$ when $2 \leq i \leq s-1$ (T_2 when $i=1$, T_{s-1} when $i=s$), select one box in the cell which is not contained in D' if it exists. Let D_i be the set of all selected boxes. Then $|D_i| \leq 3lw$. Let $D = (D' \setminus (D' \cap U_i)) \cup D_i$.

Now we show that D is an SDS of W . Note that $D' \cap U_i$ can only dominate the boxes in $W_{i-1} \cup W_i \cup W_{i+1}$ by Observation 3.8 and the boxes in $W_{i-1} \cup W_i \cup W_{i+1}$ but not in D can be dominated by D_i . Then D is a DS of W . For each $u \in W \setminus (W_{i-1} \cup W_i \cup W_{i+1} \cup D)$, then $u \notin D'$, it follows that there is $u' \in D' \cap N(u)$ such that $(D' \cup \{u\}) \setminus \{u'\} = D'_u$ is a DS of W by D' is an SDS of W . By Observation 3.8, $u' \notin U_i$, then $u' \in D \setminus D_i$. We claim $(D \cup \{u\}) \setminus \{u'\} = D_u$ is a DS of W . Since D is a DS of W , we only need to consider any $v \in N(u') \cap (W \setminus D_u)$. If $v \in W_{i-1} \cup W_i \cup W_{i+1}$, then there is $v' \in D_i$ in the cell containing v . By Observation 3.1, $v' \in N(v)$. Since $u' \notin D_i$, $u' \neq v'$. If $v \in W \setminus (W_{i-1} \cup W_i \cup W_{i+1})$, then $v \notin D'_u$ by $v \notin D_u$. Since D'_u is a DS of W , there is $v' \in D'_u \cap N(v)$. By Observation 3.8, $v' \notin U_i$, then $v' \in D \setminus D_i$. In both cases, $v' \in D_u$. Thus D_u is a DS of W .

For each $u \in (W_{i-1} \cup W_i \cup W_{i+1}) \setminus D$, there is $u' \in D_i$ such that u' and u are in the same cell. Similarly, we can show that $(D \cup \{u\}) \setminus \{u'\}$ is a DS of W . Therefore D is also an SDS of W , but $|D| < |D'|$, a contradiction with D' is a minimum SDS of W . $\square$

The algorithm will find a subset U'_i of U_i for each $1 \leq i \leq s$ such that $\bigcup_{i=1}^s U'_i$ is a minimum SDS of W . By Lemma 3.9, we only need to consider the subset of U_i with size at most $3lw$ for any $1 \leq i \leq s$. For $1 \leq i \leq s$, given sets $U'_{i+t} \subseteq U_{i+t}$ with $|U'_{i+t}| \leq 3lw$ for $t = 0, \dots, 5$. Let $\mathcal{M}_i(U'_i, \dots, U'_{i+5}) = \{U'_{i-1} \subseteq U_{i-1} : |U'_{i-1}| \leq 3lw \text{ and } U'_{i-1} \cup \dots \cup U'_{i+5} \text{ is an SDS of } W_1 \cup \dots \cup W_{i+4}\}$. Let D be a minimum SDS of $W_1 \cup \dots \cup W_{i+4}$ from $U_1 \cup \dots \cup U_{i+5}$ such that $D \cap U_{i+t} = U'_{i+t}$ for $t = 0, \dots, 5$ and $|D \cap U_j| \leq 3lw$ for $1 \leq j \leq i-1$. Define

$$m_i(U'_i, \dots, U'_{i+5}) = \begin{cases} \infty, & \text{if such } D \text{ does not exist;} \\ |D|, & \text{otherwise.} \end{cases}$$

Then $m_s(U'_s, \dots, U'_{s+5})$ is the size of a minimum SDS $D(U'_s)$ of $W_1 \cup \dots \cup W_s = W$ with $D(U'_s) \cap U_s = U'_s$. The algorithm will return a set D which has minimum cardinality among all $\{D(U'_s) : U'_s \subseteq U_s \text{ with } |U'_s| \leq 3lw\}$.

Lemma 3.10. For each $1 \leq i \leq s$ and each $U'_{i+t} \subseteq U_{i+t}$ with $|U'_{i+t}| \leq 3lw$, where $t = 0, \dots, 5$,

$$m_i(U'_i, \dots, U'_{i+5}) = \begin{cases} \infty, & \text{if } \mathcal{M}_i(U'_i, \dots, U'_{i+5}) = \emptyset; \\ \min_{U'_{i-1} \in \mathcal{M}_i(U'_i, \dots, U'_{i+5})} m_{i-1}(U'_{i-1}, \dots, U'_{i+4}) + |U'_{i+5}|, & \text{otherwise.} \end{cases}$$

Proof. If there is no $U'_{i-1} \subseteq U_{i-1}$ with $|U'_{i-1}| \leq 3lw$ such that $U'_{i-1} \cup \dots \cup U'_{i+5}$ is an SDS of $W_i \cup \dots \cup W_{i+4}$, that is, $\mathcal{M}_i(U'_i, \dots, U'_{i+5}) = \emptyset$, then $m_i(U'_i, \dots, U'_{i+5}) = \infty$.

Now we assume $\mathcal{M}_i(U'_i, \dots, U'_{i+5}) \neq \emptyset$. In the following we need to show that

$$m_i(U'_i, \dots, U'_{i+5}) = \min_{U'_{i-1} \in \mathcal{M}_i(U'_i, \dots, U'_{i+5})} m_{i-1}(U'_{i-1}, \dots, U'_{i+4}) + |U'_{i+5}|.$$

First we show that

$$m_i(U'_i, \dots, U'_{i+5}) \leq \min_{U'_{i-1} \in \mathcal{M}_i(U'_i, \dots, U'_{i+5})} m_{i-1}(U'_{i-1}, \dots, U'_{i+4}) + |U'_{i+5}|.$$

If for any $U'_{i-1} \in \mathcal{M}_i(U'_i, \dots, U'_{i+5})$, $m_{i-1}(U'_{i-1}, \dots, U'_{i+4}) = \infty$, then the above inequality clearly holds. So we assume that $m_{i-1}(U'_{i-1}, \dots, U'_{i+4}) \neq \infty$ for some $U'_{i-1} \in \mathcal{M}_i(U'_i, \dots, U'_{i+5})$. Let $U''_{i-1} = \arg \min_{U'_{i-1} \in \mathcal{M}_i(U'_i, \dots, U'_{i+5})} m_{i-1}(U'_{i-1}, \dots, U'_{i+4})$. Let D' be a minimum SDS of $W_1 \cup \dots \cup W_{i+3}$ from $U_1 \cup \dots \cup U_{i+4}$ with $D' \cap U_{i-1} = U''_{i-1}$, $D' \cap U_{i+t} = U'_{i+t}$ for $t = 0, \dots, 4$ and $|D' \cap U_j| \leq 3lw$ for $1 \leq j \leq i-2$, that is,

$$|D'| = m_{i-1}(U''_{i-1}, \dots, U'_{i+4}) = \min_{U'_{i-1} \in \mathcal{M}_i(U'_i, \dots, U'_{i+5})} m_{i-1}(U'_{i-1}, \dots, U'_{i+4}).$$

Now we show that $D = D' \cup U'_{i+5}$ is an SDS of $W_1 \cup \dots \cup W_{i+4}$ from $U_1 \cup \dots \cup U_{i+5}$. Since $D_i = U''_{i-1} \cup U'_i \cup \dots \cup U'_{i+5}$ is an SDS of $W_i \cup \dots \cup W_{i+4}$, D is a DS of $W_1 \cup \dots \cup W_{i+4}$. For any $u \in (W_1 \cup \dots \cup W_{i+1}) \setminus D$, since D' is an SDS of $W_1 \cup \dots \cup W_{i+3}$, there is $u' \in D' \cap N(u)$ such that $(D' \cup \{u'\}) \setminus \{u'\} = D'_u$ is a DS of $W_1 \cup \dots \cup W_{i+3}$. Now we show that $(D \cup \{u'\}) \setminus \{u'\} = D_u$ is a DS of $W_1 \cup \dots \cup W_{i+4}$. Otherwise there is $u'' \in (W_1 \cup \dots \cup W_{i+4}) \setminus D_u$ such that $N(u'') \cap D_u = \emptyset$, it follows that $N(u'') \cap D'_u = \emptyset$ by $D'_u \subseteq D_u$. Since D is a DS of $W_1 \cup \dots \cup W_{i+4}$, $N(u'') \cap D = \{u'\}$. Since $u \in W_1 \cup \dots \cup W_{i+1}$, by [Observation 3.8](#), $u' \in U_1 \cup \dots \cup U_{i+2}$, $u'' \in U_1 \cup \dots \cup U_{i+3}$. Further by $u'' \in W_1 \cup \dots \cup W_{i+4}$, $u'' \in W_1 \cup \dots \cup W_{i+3}$, which is a contradiction with D'_u is a DS of $W_1 \cup \dots \cup W_{i+3}$.

For any $u \in (W_{i+2} \cup W_{i+3} \cup W_{i+4}) \setminus D$, since D_i is an SDS of $W_i \cup \dots \cup W_{i+4}$, there is $u' \in D_i \cap N(u)$ such that $(D_i \cup \{u'\}) \setminus \{u'\} = D_{iu}$ is a DS of $W_i \cup \dots \cup W_{i+4}$. Now we show that $(D \cup \{u'\}) \setminus \{u'\} = D_u$ is a DS of $W_1 \cup \dots \cup W_{i+4}$. Otherwise there is $u'' \in (W_1 \cup \dots \cup W_{i+4}) \setminus D_u$ such that $N(u'') \cap D_u = \emptyset$, it follows that $N(u'') \cap D_{iu} = \emptyset$ by $D_{iu} \subseteq D_u$. Similarly, $N(u'') \cap D = \{u'\}$. Since $u \in W_{i+2} \cup W_{i+3} \cup W_{i+4}$, by [Observation 3.8](#), $u' \in U_{i+1} \cup \dots \cup U_{i+5}$, $u'' \in U_i \cup \dots \cup U_{i+5}$. Further by $u'' \in W_1 \cup \dots \cup W_{i+4}$, $u'' \in W_i \cup \dots \cup W_{i+4}$, which is a contradiction with D_{iu} is a DS of $W_i \cup \dots \cup W_{i+4}$.

Therefore D is an SDS of $W_1 \cup \dots \cup W_{i+4}$ from $U_1 \cup \dots \cup U_{i+5}$ with $D \cap U_{i+t} = U'_{i+t}$ for $t = 0, \dots, 5$ and $|D \cap U_j| \leq 3lw$ for $1 \leq j \leq i-1$. Then $m_i(U'_i, \dots, U'_{i+5}) \leq |D| = |D'| + |U'_{i+5}| = \min_{U'_{i-1} \in \mathcal{M}_i(U'_i, \dots, U'_{i+5})} m_{i-1}(U'_{i-1}, \dots, U'_{i+4}) + |U'_{i+5}|$.

Then we show that

$$m_i(U'_i, \dots, U'_{i+5}) \geq \min_{U'_{i-1} \in \mathcal{M}_i(U'_i, \dots, U'_{i+5})} m_{i-1}(U'_{i-1}, \dots, U'_{i+4}) + |U'_{i+5}|.$$

If $m_i(U'_i, \dots, U'_{i+5}) = \infty$, then the above inequality clearly holds. So we assume that $m_i(U'_i, \dots, U'_{i+5}) \neq \infty$, that is, there is a minimum SDS D of $W_1 \cup \dots \cup W_{i+4}$ from $U_1 \cup \dots \cup U_{i+5}$ with $D \cap U_{i+t} = U'_{i+t}$ for $t = 0, \dots, 5$ and $|D \cap U_j| \leq 3lw$ for $1 \leq j \leq i-1$. Then $|D| = m_i(U'_i, \dots, U'_{i+5})$. Let $D \cap U_{i-1} = U'_{i-1}$. Then $U'_{i-1} \cup \dots \cup U'_{i+5}$ is an SDS of $W_i \cup \dots \cup W_{i+4}$. Since $|U'_{i-1}| \leq 3lw$, $U'_{i-1} \in \mathcal{M}_i(U'_i, \dots, U'_{i+5})$. By [Observation 3.8](#), the boxes in U'_{i+5} cannot dominate the boxes in $W_1 \cup \dots \cup W_{i+3}$, then $D' = D \setminus U'_{i+5}$ is an SDS of $W_1 \cup \dots \cup W_{i+3}$ from $U_1 \cup \dots \cup U_{i+4}$ with $D' \cap U_{i+t} = U'_{i+t}$ for $t = -1, \dots, 4$ and $|D' \cap U_j| \leq 3lw$ for $1 \leq j \leq i-2$. Thus $\min_{U'_{i-1} \in \mathcal{M}_i(U'_i, \dots, U'_{i+5})} m_{i-1}(U'_{i-1}, \dots, U'_{i+4}) \leq m_{i-1}(U'_{i-1}, \dots, U'_{i+4}) \leq |D'| = |D| - |U'_{i+5}| = m_i(U'_i, \dots, U'_{i+5}) - |U'_{i+5}|$. $\square$

The pseudo code of the algorithm is presented in Algorithm 1 (U, W, l, w).

Theorem 3.11. Let U be a set of n unit-height boxes whose centers are inside a column T of length l and width w for some integers $l, w \geq 1$, and W be a subset of U . Algorithm 1 produces a minimum SDS D of W from U in time $O(n^{21lw})$.

Proof. By [Lemma 3.10](#), D is a minimum SDS of W from U . Now we compute the running time t of Algorithm 1. Let $|U_i| = n_i$ for any $1 \leq i \leq s$ and $n_i = 1$ for any $i = -1, 0, s+1, \dots, s+5$. Then $\sum_{i=1}^s n_i = n$. For the i th iteration, where $0 \leq i \leq s$, it takes $O((n_{i-1} \cdots n_{i+5})^{3lw})$ time. Then

$$\begin{aligned} t &= O((n_1 \cdots n_5)^{3lw}) + O((n_1 \cdots n_6)^{3lw}) + \cdots + O((n_{i-1} \cdots n_{i+5})^{3lw}) + \cdots + O((n_{s-1} n_s)^{3lw}) \\ &\leq O((n_1 + \cdots + n_5)^{3lw \times 7}) + \cdots + (n_{i-1} + \cdots + n_{i+5})^{3lw \times 7} + \cdots + (n_{s-1} + n_s)^{3lw \times 7} \\ &\leq O((n_1 + \cdots + n_5 + \cdots + n_{i-1} + \cdots + n_{i+5} + \cdots + n_{s-1} + n_s)^{3lw \times 7}) \\ &\leq O((7(n_1 + \cdots + n_s))^{3lw \times 7}) \\ &= O(n^{21lw}). \quad \square \end{aligned}$$

Algorithm 1 (U, W, l, w)**Input:** Sets U, W, U_i, W_i and integers $l, w \geq 1$ defined as above, where $0 \leq i \leq s+5$.**Output:** A minimum SDS D of W .

1. **for** each $U'_t \subseteq U_t$ with $|U'_t| \leq 3lw$, $t = 0, \dots, 5$
2. $m_0(U'_0, \dots, U'_5) = \sum_{t=1}^5 |U'_t|$;
3. **for** $i = 1$ to s
4. **for** each $U'_{i+t} \subseteq U_{i+t}$ with $|U'_{i+t}| \leq 3lw$, $t = 0, \dots, 5$
5. compute $\mathcal{M}_i(U'_i, \dots, U'_{i+5})$;
6. **if** $\mathcal{M}_i(U'_i, \dots, U'_{i+5}) = \emptyset$, **then** $m_i(U'_i, \dots, U'_{i+5}) = \infty$;
7. **else** $m_i(U'_i, \dots, U'_{i+5}) = \min_{U'_{i-1} \in \mathcal{M}_i(U'_i, \dots, U'_{i+5})} m_{i-1}(U'_{i-1}, \dots, U'_{i+4}) + |U'_{i+5}|$,
8. $M_i(U'_i, \dots, U'_{i+5}) = \arg \min_{U'_{i-1} \in \mathcal{M}_i(U'_i, \dots, U'_{i+5})} m_{i-1}(U'_{i-1}, \dots, U'_{i+4})$;
9. $U'_s = \arg \min_{U'_s \subseteq U_s, |U'_s| \leq 3lw} m_s(U'_s, \dots, U'_{s+5})$;
10. $D \leftarrow \emptyset$;
11. **for** $i = s$ to 1
12. $D \leftarrow D \cup U'_i$;
13. $U'_{i-1} \leftarrow M_i(U'_i, \dots, U'_{i+5})$;
14. **Return** D .

When $W = U$, Algorithm 1 produces a minimum SDS of U in $O(n^{21lw})$ time. Algorithm 1 is also effective when U is a set of unit cubes. Hence we have the following result.

Corollary 3.12. *Let U be a set of n unit cubes whose centers are inside a column T of length l and width w for some integers $l, w \geq 1$, and W be a subset of U . Algorithm 1 produces a minimum SDS of W from U in time $O(n^{21lw})$.*

When P is a set of n unit balls, the result of [Observation 3.8](#) is also valid. Replace the cell of size $1 \times 1 \times 1$ in [Lemma 3.9](#) with the cell of size $\frac{1}{\sqrt{3}} \times \frac{1}{\sqrt{3}} \times \frac{1}{\sqrt{3}}$. Now any two unit balls with centers in a cell intersect. Then for any minimum SDS D of W from U , $|D \cap U_i| \leq 9\sqrt{3}lw$ for any $1 \leq i \leq s$. In this way, by replacing “ $3lw$ ” with “ $9\sqrt{3}lw$ ” in the discussion about the unit-height box above, the following result can be obtained.

Corollary 3.13. *Let U be a set of n unit balls whose centers are inside a column T of length l and width w for some integers $l, w \geq 1$, and W be a subset of U . Algorithm 1 produces a minimum SDS of W from U in time $O(n^{63\sqrt{3}lw})$.*

When P is a set of n unit-height rectangles, or n unit squares, or n unit disks, the result of [Observation 3.8](#) is also valid. Now assume that the centers of these objects are inside a strip T of width w , where $w \geq 1$ is an integer. We use horizontal line to partition T into $(w, 1)$ -rectangles, which are rectangles with size $w \times 1$. For the case of unit-height rectangle or unit square, replace the cell of size $1 \times 1 \times 1$ in [Lemma 3.9](#) with the cell of size 1×1 . Note that any two unit-height rectangles or any two unit squares with centers in a cell intersect. Then for any minimum SDS D of W from U , $|D \cap U_i| \leq 3w$ for any $1 \leq i \leq s$. In this way, by replacing “ $3lw$ ” with “ $3w$ ” in the discussion about the unit-height box above, the following result can be obtained.

Corollary 3.14. *Let U be a set of n unit-height rectangles or n unit squares whose centers are inside a strip T of width w for some integer $w \geq 1$, and W be a subset of U . Algorithm 1 produces a minimum SDS of W from U in time $O(n^{21w})$.*

For the case of unit disk, replace the cell of size $1 \times 1 \times 1$ in [Lemma 3.9](#) with the cell of size $\frac{1}{\sqrt{2}} \times \frac{1}{\sqrt{2}}$. Note that any two unit disks with centers in a cell intersect. Then for any minimum SDS D of W from U , $|D \cap U_i| \leq 6w$ for any $1 \leq i \leq s$. In this way, by replacing “ $3lw$ ” with “ $6w$ ” in the discussion about the unit-height box above, the following result can be obtained.

Corollary 3.15. *Let U be a set of n unit disks whose centers are inside a strip T of width w for some integer $w \geq 1$, and W be a subset of U . Algorithm 1 produces a minimum SDS of W from U in time $O(n^{42w})$.*

3.3. 2-Level shifting strategy

In this subsection, we apply 2-level shifting strategy and the above dynamic programming approach to give PTAS for the MSDS problem. We also only discuss the case of unit cube graphs, the other graph classes are similar.

The basic approach is similar to the 3-level shifting strategy. The 2 levels are exactly the first two levels of 3-level shifting strategy, and the third level is missing. For convenience, we use the same symbols as 3-level shifting strategy if there is no specifically stated. Since the third level is missing, the final box C has size $6k \times 6k \times H$ and C^d has size $(6k + 2d) \times (6k + 2d) \times H$. Thus C^d is a column. Then applying Algorithm 1 ($U_{C^3}, U_{C^2}, 6k + 6, 6k + 6$) can obtain a minimum SDS OPT_{C^2} of U_{C^2} from U_{C^3} , which runs $O(n_{C^3}^{756(k+1)^2}) = n_{C^3}^{O(k^2)}$ time, where $n_{C^3} = |U_{C^3}|$.

In the following we consider any (i, j) -th iteration, where $1 \leq i, j \leq k$. Suppose there are s boxes $C_1, \dots, C_s$ in this partition. Let $S_{i,j} = OPT_{C_1^2} \cup \dots \cup OPT_{C_s^2}$. The algorithm outputs a set S , which has minimum cardinality among all $\{S_{i,j} : 1 \leq i, j \leq k\}$, that is, $|S| = \min_{1 \leq i, j \leq k} |S_{i,j}|$.

Lemma 3.16. S is an SDS of U .

Proof. The proof is the same as Lemma 3.3. $\square$

Lemma 3.17. $|S| \leq (1 + \frac{6}{k})|OPT|$.

Proof. Consider any (i, j) -th iteration, where $1 \leq i, j \leq k$. Same as Lemma 3.4, we have

$$\sum_{i=1}^k |OPT_i^Y| = |\bigcup_{i=1}^k OPT_i^Y| \leq |OPT|. \quad (3.7)$$

$$\sum_{j=1}^k |OPT_j^X| = |\bigcup_{j=1}^k OPT_j^X| \leq |OPT|. \quad (3.8)$$

Let $OPT_{i,j} = OPT_i^Y \cup OPT_j^X$. Then by inequalities (3.7) and (3.8), we have

$$\sum_{i=1}^k \sum_{j=1}^k |OPT_{i,j}| \leq k \sum_{i=1}^k |OPT_i^Y| + k \sum_{j=1}^k |OPT_j^X| \leq 2k|OPT|. \quad (3.9)$$

Thus

$$\begin{aligned} |S_{i,j}| &\leq |OPT_{C_1^2}| + \dots + |OPT_{C_s^2}| \\ &\leq |OPT \cap U_{C_1^3}| + \dots + |OPT \cap U_{C_s^3}| \\ &\leq |OPT| + 3|OPT_{i,j}|. \end{aligned} \quad (3.10)$$

The second inequality is obtained by the same reason as Lemma 3.4. The third inequality is obtained by the fact that only the cubes in $OPT_{i,j}$ can be repetitively calculated and each cube is repetitively calculated at most 4 times.

Therefore by inequalities (3.9) and (3.10) we have

$$\begin{aligned} |S| &= \min_{1 \leq i, j \leq k} |S_{i,j}| \\ &\leq \frac{1}{k^2} \sum_{i=1}^k \sum_{j=1}^k |S_{i,j}| \\ &\leq \frac{1}{k^2} (k^2|OPT| + 3 \sum_{i=1}^k \sum_{j=1}^k |OPT_{i,j}|) \\ &\leq \frac{1}{k^2} (k^2|OPT| + 6k|OPT|) \\ &= (1 + \frac{6}{k})|OPT|. \quad \square \end{aligned}$$

Theorem 3.18. Given a set U of n unit cubes and an integer $k \geq 1$. The proposed 2-level shifting algorithm produces an SDS of U with size at most $(1 + \frac{6}{k})|OPT|$ in time $n^{O(k^2)}$.

Proof. By Lemmas 3.16, 3.17, the algorithm produces an SDS of U with size at most $(1 + \frac{6}{k})|OPT|$. Now we discuss the running time. For each (i, j) -th iteration ($1 \leq i, j \leq k$), the time taken to compute OPT_{C^2} is $n_{C^3}^{O(k^2)}$, where C is any one of the boxes in this iteration and $n_{C^3} = |U_{C^3}|$. Since each cube participates in computing $S_{i,j}$ at most 4 boxes, computing $S_{i,j}$ needs $n^{O(k^2)}$ time. There are k^2 iterations, the total running time is $n^{O(k^2)}$. $\square$

When U is a set of unit balls, the 2-level shifting algorithm is also effective. Hence we have the following result.

Corollary 3.19. *Given a set U of n unit balls and an integer $k \geq 1$. The proposed 2-level shifting algorithm produces an SDS of U with size at most $(1 + \frac{6}{k})|OPT|$ in time $n^{O(k^2)}$.*

Similarly, for 2-dimensional situation, 2-level shifting strategy can be reduced to 1 level. Thus we have the following result.

Theorem 3.20. *Let U be a set of n unit squares or n unit disks and $k \geq 1$ be an integer. The algorithm outputs an SDS of U with size at most $(1 + \frac{1}{k})|OPT|$ in time $n^{O(k)}$.*

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Data availability

No data was used for the research described in the article.

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